The small anchored imprimitive endpoint

Human Compression project: bounded L044 proof component

The accepted anchored imprimitive bound covers degree at least twelve. This note closes its smaller degrees, using the same actual-orbit forest and the already paid maximal symmetric-string interfaces. It supplies only the anchored replacement needed in L044's supported interval induction; it does not restore the false unrestricted reading of its Proposition 5.13. An endpoint anchor is a point moved by that endpoint generator and fixed by every other generator.

Lemma 1 (Small symmetric endpoint test)**.** *Let $(h_0, \dots, h_{b-2})$ be a connected string C-marking of S_b of length $b-1$, $2 \le b \le 5$. Put $Q = \langle h_1, \dots, h_{b-2} \rangle$ and $F = \langle h_0, \dots, h_{b-3} \rangle$. If F fixes a point p, then*

$$C_{S_b}(Q) \cap (S_b)_p = 1.$$

Moreover an even-weight binary vector constant on the Q-orbits and zero at p is zero.

Proof. The paid degree-five maximal-string certificate gives the ordinary adjacent-transposition marking. For $b = 2, 3$ this is elementary: distinct involutions of S_3 are transpositions. For $b = 4$, the accepted elementary classification gives the ordinary path and, up to reversal and relabelling,

$$((12)(34), (23), (34)).$$

Here is the short classification argument. Both commuting end involutions cannot be double transpositions: adjoining one involution to the normal four-group does not generate S_4. For two disjoint single transpositions, the middle must be a single transposition joining their pairs; a double transposition preserves that pair partition. For one single and one double end, they share one factor, and again the middle must join their pairs by a single transposition. These give exactly the two forms just displayed.

For $b = 2$, a point stabilizer is trivial and an even binary vector zero at one of the two points is zero. For $b \ge 3$ in an ordinary path labelled $(12), (23), \dots$, the only F-fixed point is $p = b$. The group $Q = S_{\{2,\dots,b\}}$ has trivial centralizer fixing p: for $b \ge 4$ its centralizer in S_b is trivial; for $b = 3$ the possible nonidentity element swaps p. Its large orbit contains p, so a binary vector as stated can be nonzero only at point 1, which has odd weight.

In the displayed exceptional order, F is transitive D_8, so there is no p. In the reversed order, F is a point stabilizer S_3, whereas Q is transitive D_8. An element centralizing a transitive group and fixing a point fixes every point. A Q-constant vector zero at one point is likewise zero. □

Theorem 2 (Small anchored imprimitive bound)**.** *Let $H \le A_N$, $4 \le N \le 11$, be transitive and imprimitive, with a connected string C-marking of rank d, all maximal omissions intransitive, and an endpoint anchor. Then*

$$d \le \left\lfloor \frac{N-2}{2} \right\rfloor.$$

Proof. Choose blocks of size a and number b. Let M be a minimal distinguished subfamily generating the whole block action, and let c be its actual number of point orbits. The accepted forest argument and Whiston bound give

$$d - |M| \le c - 1, \qquad c \le a, \qquad |M| \le b - 1.$$

Thus $d \le a + b - 2$. If $a, b \ge 3$, the only case here is $a = b = 3$. Equality $d = 4$ would force $|M| = 2, c = 3$ and block image S_3. Its three orbits each meet each block once, so $\langle M \rangle$ is faithfully the diagonal S_3 on three copies. A block transposition lifts to three point transpositions, contradicting evenness. Hence $d \le 3$ as required.

Suppose $b = 2$. Reverse the marking so an anchor p is moved only by its first generator τ. Every other generator fixes p, hence preserves both blocks; transitivity forces τ to interchange them. It is a product of a transpositions, so a is even. Only $a = 2, 4$ remain. To exclude $d = a$, normalize the two blocks as row pairs $1, \dots, a$ and $1', \dots, a'$ with $p = 1$ and $\tau = \prod_j (j\ j')$. Put $x = s_1$.

For $a = 2$, x fixes 1 and must move $1'$ to connect its row pair to the other pair. Thus $x = (1'\ 2')$ is odd. For $a = 4$, the last two generators centralize τ, fix the anchor row, and are independent. Their diagonal row action is the full S_3, so write them

$$h_0 = (23)(2'3'), \qquad h_1 = (34)(3'4').$$

The generator x centralizes h_1, fixes 1 and preserves the two blocks. It must move $1'$; consequently

$$x = (1'2')(34)^{\epsilon}(3'4')^{\delta}, \qquad \epsilon + \delta \equiv 1 \pmod 2.$$

But $\langle \tau, x, h_0 \rangle$ is transitive: τ joins the two points of each row, while the other edges connect row 1 to 2, row 2 to 3, and row 3 to 4. This is the forbidden transitive omission of h_1. Neither rank-a case is possible.

Finally let $a = 2$, so $2 \le b \le 5$, and suppose $d = b$. All three inequalities are equalities: M has $b - 1$ members, its block image is S_b, and its two actual orbits are natural copies of that block action. Thus $\langle M \rangle$ is faithfully diagonal S_b with its inherited string IP. It cannot split into nontrivial commuting marked factors: S_b is directly indecomposable for $2 \le b \le 5$ (its elementary normal-subgroup lists give this immediately). Hence the members of M occupy one interval. The remaining generator x is an end; reverse so it is first.

An anchor cannot be at x, since $\langle M \rangle$ has no fixed point. It must be at the far endpoint of M. Its block p is fixed by that endpoint's omission in M, and x fixes the anchor point. The block image of x centralizes the near-end omission Q in M and fixes p. Lemma 1 makes it trivial. Thus x is a binary vector of flips of the two-point blocks. String commutation makes this vector Q-constant, the anchor makes its coordinate at p zero, and evenness makes its weight even. The same lemma makes the whole vector zero, contradicting the nonidentity generator. This excludes $d = b$ and finishes every small block size. □

Exact scope. Together with the accepted degree-at-least-twelve theorem, the anchored bound now holds in every imprimitive degree. Only the exact endpoint anchor and intransitive omissions make the small cases work: parity and string commutation alone do not eliminate the two degree-eight candidates, but their last omissions are transitive. No finite trial is used as a proof premise. The already accepted S_5 maximal-string certificate and full forest/common proof remain paid dependencies. No new whole-source or formal claim is made.